



CONTACT

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- Third Year - IT Student
- CGPI (T.E.): 8.40

EDUCATION

B.E. Information Technology

St. Francis Institute of Technolgy
2023-present

Higher Education

Ryan International School
2021-2023

Junior Education

Ryan International School
2009-2021

SKILLS

- Project Management
- Adaptability
- Teamwork
- Time Management
- Leadership
- Effective Communication
- Critical Thinking

LANGUAGES

- English
- Hindi
- Kannada

KARTHIK JANARDHAN

THE FOLLOWING ARE MY EXPERIENCES:

FlyRank AI - Internship

January 2026 - Present

Machine Learning Engineering Intern

- Wrangled messy search and analytics data end to end, with deliberate handling of joins, missing values and nested fields to build reliable modelling datasets.
- Applied semantic methods — embeddings, clustering and classification — to a focused search-quality problem.
- Delivered 3-5 non-obvious insights from the analysis, each backed by direct evidence from the model output.

St. Francis Institute of Technology

15th September 2025 - 11th January 2026

Research & Development Engineer

- Leading development of an internal academic web platform under the college R&D initiative, owning full-stack architecture and delivery.
- Building modern frontend interfaces with Next.js, React, TypeScript and Tailwind CSS, and backend APIs with FastAPI and PostgreSQL.
- Implementing authentication, validation and data management systems for secure, scalable academic workflow automation.
- Assigning development tasks, coordinating the team, and overseeing system design, deployment workflows and performance optimisation.

NSE(National Stock Exchange) - Internship

1st June 2026 - 1st July 2026

Application Developer - WEBTECH

- Designed and implemented an observability stack for monitoring LLM serving infrastructure, with real-time tracking of token consumption, model failures, request latency, and GPU utilization. Used Prometheus for metric scraping and alerts, Grafana for dashboards, DCGM Exporter for NVIDIA GPU telemetry, and LiteLLM + Langfuse for tracing and token-level visibility. Enabled faster failure investigation, better load correlation, and more informed capacity planning across model-serving workloads.
- The system helped identify failed models, monitor resource utilization, correlate incidents with GPU pressure, and improve capacity planning and reliability of model-serving workloads

Google Cloud - Internship

15th November 2025 - 5th January 2026

Intern

- Completed a Virtual Internship on Google Cloud Generative AI conducted by SmartBridge in collaboration with AICTE (Nov 2025 - Jan 2026).
- Gained hands-on experience in Generative AI concepts, real-world use cases, and practical implementation on Google Cloud.
- Built a strong foundation in applying AI-driven solutions through guided learning and practical exercises.

Google Cloud - Gen Ai

January - June 2025

Leading the team

- Successfully completed Google Cloud GenAI Academy, gaining hands-on experience with Generative AI tools and workflows.
- Acquired practical skills in Vertex AI, Gemini APIs, Imagen, Streamlit, and Multimodal RAG through industry-aligned coursework.
- Recognized as a Top 50 Learner (June) among 250,000+ participants and ranked in the Top 1 performers, earning 800 points.

Colloquium

March 2024

Winner and leader

- Won First Prize at Colloquium'25 Inter-College IT Hackathon themed "IT for Good Governance" for the project Itihaas.
- Co-developed a heritage tourism platform enabling guided tours, interactive maps, rewards, and curated merchandise for enhanced user engagement.
- Contributed to solution design focused on digital governance, cultural preservation, and tourism technology through collaborative teamwork.

Intern

- Completed Mastery Level Skill Training Internship at GET AnalyticX, gaining hands-on experience in AI tools, Python, and Django-based application development.
- Worked with GitHub, web scraping, and chatbot development, strengthening practical software and automation skills.
- Designed and delivered a live Power BI project (IPL Dashboard), applying data visualization techniques to derive actionable insights.

SOFTWARE PROJECTS

1. CyberQore - Multi-Agent Explainable SOC Copilot (LangGraph, ChromaDB, Llama 3.1) June 2026 - Present

- Retrieval-grounded, 8-layer open-source SOC platform that turns raw alerts into cited investigations, orchestrating seven specialist agents under a LangGraph Supervisor (Drain3 → DeepLog LSTM, pySigma → MITRE ATT&CK, LightGBM + YARA on EMBER/BODMAS, XGBoost + DistilBERT phishing, incident response, reporting, EPSS/CISA KEV risk) — with prompt-injection defence treating every log line as untrusted, SHAP TreeExplainer verdicts grounded by inline RAG citation IDs, a 0–100 trust score via NLI entailment (HHEM-2.1) and independent agent cross-check; built on BGE-M3, ChromaDB, bge-reranker and Llama 3.1 8B via Ollama.

2. Production-Grade Cyclic RAG - LangGraph, Guardrails & LLM Gateway (GCP, Terraform) December 2025 - March 2026

- Enterprise RAG system that separates True Data from Noisy Data using cross-encoder re-ranking and self-corrects through a cyclic LangGraph (query-transform → retrieve loop plus a regeneration edge, bounded so it always terminates) — hybrid tsvector + vector retrieval via RRF with MMR over pgvector, a LiteLLM gateway for multi-provider routing, fallbacks and cost budgets, full input/output guardrails, and a RAGAS eval gate in Cloud Build CI with Terraform IaC on Cloud Run; hybrid retrieval lifted context precision ~0.72 → ~0.81 and recall ~0.68 → ~0.79 over vector-only.

3. Enterprise Advanced RAG - Kubernetes IT-Ops Assistant (FastAPI, Qdrant, Redis) March - June 2026

- Production RAG platform for Kubernetes IT operations, built as a Postgres-checkpointed LangGraph state machine routing across RAG, Text2SQL and hybrid paths — HyDE expansion, dense + BM25 hybrid retrieval with RRF (k=60), BGE/Voyage cross-encoder re-ranking, a CRAG grader falling back to Tavily web search below 0.7 relevance, Spotlighting chunk isolation and Self-RAG reflection, with Text2SQL running SELECT-only validation behind a LangGraph interrupt() human-in-the-loop gate; hardened by a 9-layer defense-in-depth guardrails pipeline and 5-tier Redis cache on AWS ECS Fargate.

4. NSE_Observability - LLM Serving Observability (Prometheus, Grafana, Langfuse) June - July 2026

- The system helped identify failed models, monitor resource utilization, correlate incidents with GPU pressure, and improve capacity planning and reliability of model-serving workloads. Observability stack for LLM serving infrastructure using Prometheus, Grafana, DCGM Exporter, LiteLLM and Langfuse to monitor latency, token usage, GPU utilization and model failures — improving troubleshooting and capacity planning.

5. EV-Bharat - Intelligent EV Ecosystem (Python, FastAPI, Scikit-learn, YOLO) March - April 2026

- AI-integrated microservice platform bridging EV hardware (ESP32/BMS telemetry) with predictive intelligence. Trained RandomForest and GradientBoosting regressors for battery State-of-Charge, State-of-Health and Remaining-Useful-Life prediction; added a MediaPipe computer-vision driver drowsiness engine, a YOLO object-detection service, and a Gemini-powered RAG advisory bot for Indian EV compliance.

6. Failure-Aware Search Agents - DPO-Aligned Agents (TRL, PEFT/QLoRA, Google ADK, MCP) April - May 2026

- Post-trained LLM search agents with Direct Preference Optimization so they express calibrated uncertainty and trigger re-search instead of hallucinating confidently — ambiguous-query datasets (HotpotQA/TriviaQA/AmbigQA), k=8 ReAct rollouts and a composite reward (0.5-BERTScore + 0.3-calibration + 0.2-recovery) scored into DPO pairs for QLoRA fine-tuning of Mistral-7B-Instruct, served live through a Google ADK SequentialAgent over a FastMCP tool server that routes each query to confident, hedge or re-search.

TECHNICAL AND SOFT SKILLS

7. CrimeIntel AI - Crime Intelligence & Geospatial Analytics (Next.js, Node.js, NLP) July 2026

- Real-time crime intelligence platform turning heterogeneous news, social and open-government feeds into actionable intelligence. Built an NLP pipeline for crime-type classification (9+ categories), severity scoring and NER-based entity extraction, followed by geocoding enrichment, cross-source deduplication and Kernel Density Estimation hotspot heatmaps rendered live over WebSockets.

8. HealthSaathi - Patient Pre-Consultation Assistant (Node.js, PostgreSQL, AssemblyAI) April 2025

- Clinical intake assistant that captures full symptom context and generates doctor-ready reports. Implemented smart pattern detection over report history, rule-based urgency triage, cycle-correlated symptom insight, medical-jargon simplification, and AssemblyAI speech-to-text voice intake with multilingual and offline-first support.

9. VendorConnect - Local Street Food System (Python, OpenAI, Android) September 2025 - January 2026

- Digital platform solving raw-material sourcing for Indian street food vendors — connecting vendors with verified suppliers, enabling group ordering for bulk discounts and real-time price alerts, with a multilingual AI assistant.

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- Languages :** Python, C, C++, Java, JavaScript, TypeScript, SQL
 - AI / ML :** PyTorch, Scikit-learn, Hugging Face, XGBoost, LightGBM, DistilBERT, Pandas, NumPy, OpenCV, YOLO, MediaPipe, Classification, Regression, Clustering, NLP & NER, SHAP Explainability, DPO / QLoRA / PEFT Fine-Tuning
 - LLM & Agentic Systems :** LangGraph, LangChain, Multi-Agent Orchestration, Google ADK, MCP, RAG (Hybrid Search, RRF, MMR, HyDE, CRAG, Self-RAG, Re-ranking), Text2SQL, LLM Guardrails, LLM Gateways (LiteLLM), Qdrant, ChromaDB, pgvector, RAGAS Evaluation, Ollama / Llama 3.1, Vertex AI, Gemini API, Prompt Engineering
 - Data & MLOps / Cloud :** Data Wrangling & Feature Engineering, Power BI, Prometheus, Grafana, Langfuse, LangSmith, OpenTelemetry, Docker, Terraform, GCP (Cloud Run, Cloud Build, Eventarc), AWS ECS, GitHub Actions
 - Web Technologies :** HTML, CSS, JavaScript (ES6+), React.js, Next.js, Node.js, Express.js, FastAPI, Django, Flask, RESTful APIs, JSON, Tailwind CSS
 - Databases :** PostgreSQL, MySQL, MongoDB, Redis, SQLite, Supabase
 - Tools :** VS Code, Git, GitHub, Linux (Ubuntu), Kali Linux, Docker, MS Office
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- Internship Completion Certificate - GET AnalyticX
 - Certificate of Completion - Gen AI Program - Gen AI Academy (Forge)
 - Virtual Internship Completion Certificate - Generative AI - SmartBridge
 - Introduction to Technology Apprenticeship - Job Simulation - Accenture (Forage)
 - Cybersecurity Analyst - Job Simulation - Accenture (Forage)
 - ISRO Course - Government of India
 - Introduction to Generative AI - IBM SkillsBuild
 - Certificate of Appreciation - India's Biggest GenAI Global AI Event - NXT WAVE / OpenAI Academy
 - Certificate of Acknowledgement - Bharatiya Antariksh Hackathon
 - Google Developer Groups on Campus
 - NoSQL & MongoDB Bootcamp - DevTown
 - Microsoft Student Chapter (MSIT)
 - Google Cloud Generative AI / AI Completion Certificate - Google Cloud
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- GitHub profile : <https://github.com/karthikj30>

CERTIFICATES

LINKS

REFERENCE

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